RxLens – Prescription Reader

CELESTIA | 24 HR NATIONAL HACKATHON | Open Innovation

Prescription photo → clear, plain-language medicine plan.

Team Name: Icarus

Members: Garvaansh Soni (Team Lead), Ayansh Srivastava, Kushagra Bennuri

College: JSS Science and Technology University, Mysuru

01 Problem Statement

What problem are you solving?

THE PROBLEM

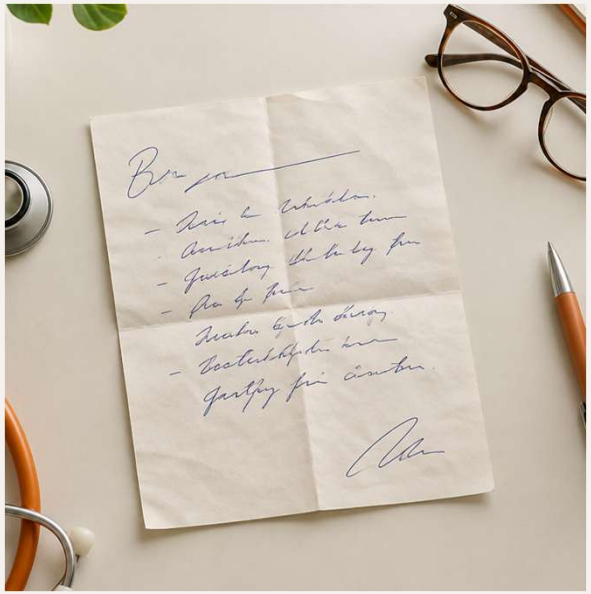
Handwritten prescriptions are hard to read, and doses and timings are rarely explained in plain language.

WHO FACES IT

Elders, rural and low-literacy patients, and caregivers managing someone else's medicines.

WHY IT MATTERS NOW

Missed or wrong doses slow recovery and cause avoidable harm. Modern OCR and vision models make a fast fix possible.



02 Objective

What does your solution aim to achieve?

Goal: turn a prescription photo into a clear, plain-language medicine plan.

85%+

Name accuracy
Medicine names read correctly
Measured on a set of test prescriptions

EN

Plain language
Simple English explanations
Dose, purpose and cautions per medicine

<15s

End-to-end speed
Photo to schedule
Timed from upload to finished plan

PROPOSED TARGETS — NOT VERIFIED RESULTS.

03 Proposed Solution

Your idea in simple words

- Build: a web app that reads a prescription photo and returns a verified, easy-to-follow medicine plan.
- How it helps: every medicine is checked against a drug database, explained simply and turned into reminders.

AI READING

Extracts medicine names, doses, frequency and duration from the photo.

DRUG-DB CHECK

Matches names to a public drug dataset, fixes OCR errors, flags low confidence.

PLAIN-LANGUAGE GUIDE

Explains purpose, how to take it and cautions in simple English.

SMART REMINDERS

Builds a daily schedule with in-app reminders and a caregiver view.

04 Innovation & Uniqueness

What makes it different?

EXISTING: GENERIC OCR APPS
ILLUSTRATIVE RAW OUTPUT

```
Tab — ? mg
Morning / night?
Name uncertain
```

Gaps: raw text only, no verification, no advice.

RxLens

READ · VERIFY · EXPLAIN

Medicine name: 92% confidence
Dose & timing: Review if unclear
Purpose: Plain language
Reminders: Caregiver view

Sample confidence and medicine data are illustrative, not validated.

WHAT IS NEW IN OUR APPROACH

- **Verifies against a drug database, not just OCR**
- **Explains doses and cautions in plain language**
- **Reminders with a caregiver view**

05 Tech Stack

Tools and technologies you will use

AI / ML

- **EasyOCR / vision LLM**
  reads printed and handwritten text
- **Fuzzy name matching**
  corrects OCR spelling errors
- **Confidence scores**
  flag unclear fields for review

BACKEND & DATA

- **Python · Flask / FastAPI**
  REST API and pipeline logic
- **PostgreSQL / MongoDB**
  profiles and saved schedules
- **Public drug dataset**
  names, uses and cautions

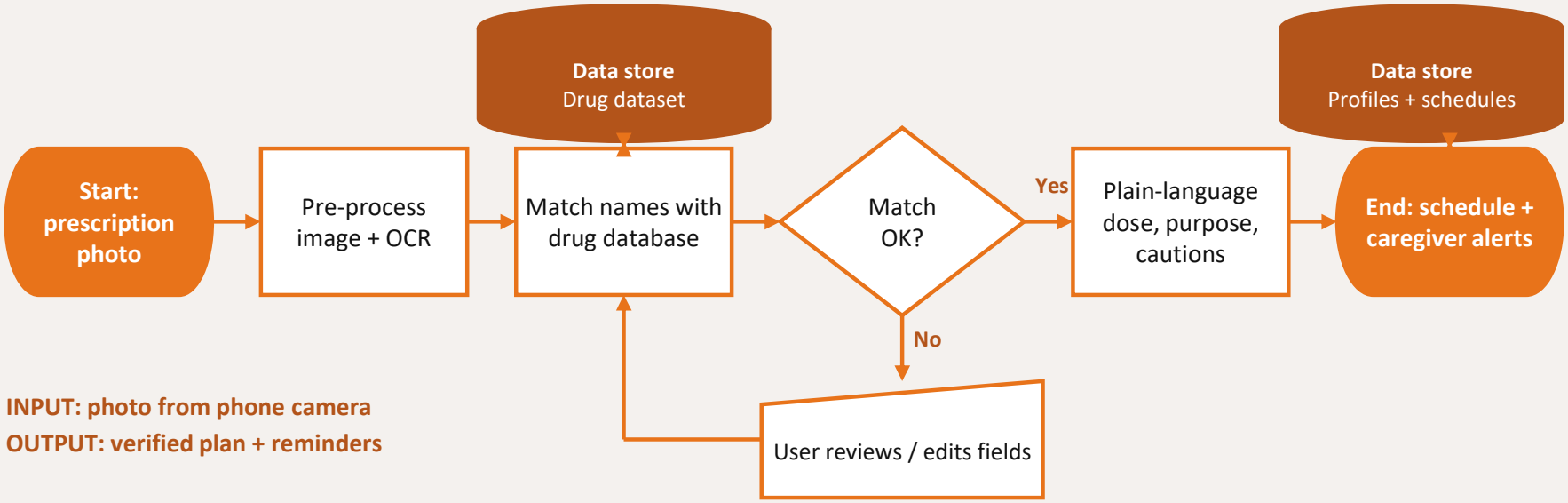
WEB & CLOUD

- **React web app**
  responsive, phone-friendly photo upload
- **Render / Vercel**
  quick cloud hosting

Selected components remain interchangeable during the sprint. Hardware: none needed, runs on any phone browser.

06 Architecture / Workflow

How the system works



07 Feasibility

Can it be built in 24 hours?

0–6 h
OCR + DB

Photo upload, text extraction, drug dataset loaded

6–16 h
App build

API, React UI, explanations and reminders

16–24 h
Polish + demo

Test on sample prescriptions, fix bugs, rehearse demo

TEAM

- Garvaansh: ML + OCR
- Ayansh: full-stack web
- Kushagra: database + backend

RESOURCES

- Pretrained OCR / vision LLM
- Public drug dataset
- Render / Vercel hosting

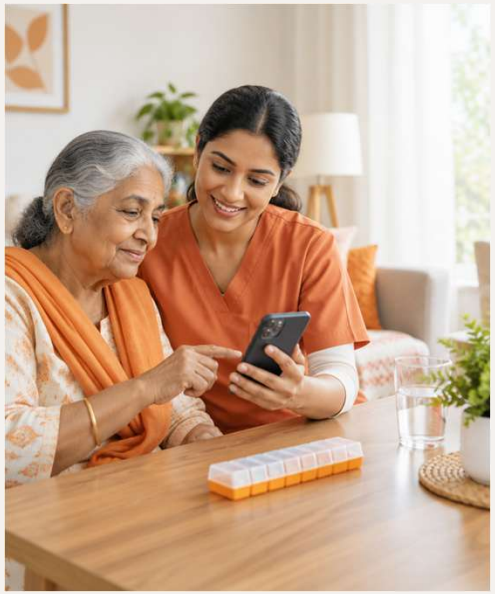
RISKS → PLAN

- Messy handwriting → confidence score + manual edit
- OCR typos → fuzzy match with drug DB

Scope is fixed to one end-to-end demo flow. Informational only, not medical advice.

08 Impact

Who benefits and how



USERS

- Elderly patients
- Caregivers
- Rural, low-literacy families

SOCIAL IMPACT

- Safer medication use
- Better adherence
- Fewer missed doses
- Informed patients

BUSINESS VALUE

Pharmacy integrations · Clinic + telehealth links · Refill partnerships

SCALABILITY Cloud API · more users · new drug data

09 Future Scope

Where can this go next?

ALL ITEMS BELOW ARE FUTURE PLANS

NEXT FEATURES

- Drug-interaction warnings
- Allergy warnings
- Dose history and refill tracking

PARTNERSHIP OPPORTUNITIES

- Pharmacy partnerships
- Telemedicine partnerships
- Easy refills and delivery
- Clinic and hospital onboarding

LONG-TERM VISION

“A medication companion for every patient.”

From reading prescriptions to guiding the whole treatment journey.

10 Team Details

Who is behind the idea

Member 1 (Team Lead)

Garvaansh Soni

ML Engineer

Responsibilities

- OCR / vision-LLM integration
- Fuzzy name matching
- Model tuning + confidence scores

Member 2

Ayansh Srivastava

Full-Stack Developer

Responsibilities

- React web app and photo upload
- API integration
- App Reminder flow

Member 3

Kushagra Bennuri

Database & Backend

Responsibilities

- Drug dataset + database
- User profiles + schedules
- Backend API